Tower up: genus ladder

Tower up: one elliptic discrete log solved at genus 1, 2 and 3

Status: construction + measurement, same characteristic. A boundary, not an improvement. Every number is an unverified assertion `[pcc:unverified]` (no independent validator has replayed it). Follow-up to PR #3898 (`../ATLAS.md`, `../RESULTS.md`).

Question

The transfer atlas showed that a degree-2 cover never moves the log to its complementary factor. The other direction remains: go **up** the tower. Carry the log into higher-genus Jacobians and ask whether the destination is easier.

Construction (`src/cryptanalysis/chained_transfer/tower.rs`)

For a cubic model `E : Y² = c(s)` and `a ∈ F_p^*`:

```
F(X) = c3·Y_ + c2·Y_² + c1·Y_³ + c0·Y_⁴,   Y_ = X − a²,    C : y² = F(x²)
φ : C → E,   (x, y) ↦ (s, Y) = (1/(x² − a²), y·s²)       (F(x²) = c(s)/s⁴)
```

- `C` has genus 3 when `c0 ≠ 0`, and genus 2 when `c0 = 0 ≠ c1`, that is, when E has a 2-torsion point at `s = 0`. So the **same** curve E covers both a genus-2 and a genus-3 curve: the genus-2 cover uses the model translated to a rational 2-torsion point, the genus-3 cover a model with `c(0) ≠ 0`.
- The Jacobian is the repository's word-size Cantor group `HCurve`, on the odd model with `(a, 0)` sent to infinity.
- The pull-back is explicit: `φ^*([Q]−[O]) = [P₊−∞] + [P₋−∞] + [W−∞]`, with `W = (−a, 0)`. This holds because `φ^*(O) = ∞ + W` and `[W − ∞]` is 2-torsion.
- The one carried log is checked as follows. `ord φ^*(P) = r` is checked. `φ^*([k]P) = [k]φ^*(P)` is checked on every multiple `k < 80` whose preimages are rational: 3–35 checks per rung. Every solve recovers the planted log.
- Unit test: `cargo test --release --lib chained_transfer::tower` passes.

Measurement (`tower_bench`)

Eight primes `p ∈ {53 … 251}` (the `HCurve` range). One instance per prime, [pcc:unverified] 8 seeds per cell, counted group operations, all solves verified.

p	r	rho on E	rho on $J(C_2)$	rho on $J(C_3)$	IC on $J(C_2)$	IC on $J(C_3)$	IC_3 / IC_2
53	11	131	132	134	70	170	2.42
79	11	137	132	129	106	298	2.82
101	19	185	177	180	126	293	2.33
131	67	337	314	305	148	503	3.40
151	19	181	181	168	181	580	3.20
181	41	249	276	255	210	684	3.25
211	31	224	220	229	234	789	3.37
251	37	248	270	251	267	829	3.10

The IC columns count Jacobian group operations only. `hyper_ic` leaves its sparse linear algebra over `Z/r` and its smoothness oracle unpriced. Priced per AGENTS.md §5a, at 0, 0.1 and 1 GAE per linear-algebra multiplication, IC divided by rho on E is:

p	IC_2 @0	@0.1	@1	IC_3 @0	@0.1	@1
53	0.53	2.09	16.1	1.29	4.73	35.7
101	0.68	4.25	36.4	1.58	12.35	109.3
151	1.00	7.19	62.9	3.20	24.21	213.3
251	1.08	10.97	100.0	3.34	35.82	328.1

The smoothness-oracle field multiplications are left out of this table too. Including them could only raise the IC figures.

The figure `tower.svg` plots the group-operation columns.

What the run shows

1. **Going up the tower makes the solve harder.** IC on genus 3 costs **2.3–3.4×** IC on genus 2 at every p. This is about the 3× that the smoothness probability `1/g!` predicts (`3!/2! = 3`).
2. **Genus 3 is above rho at every price.** It costs at least 1.29× rho on E even when the linear algebra is free.
3. **Genus 2 cannot be ranked against rho at this size.** At 0 GAE its group operations alone are 0.44–1.08× rho, so the order does not hold across the §5a band. At 0.1 and 1 GAE it is above rho (2.1–100×). Two things depress the 0-GAE figure: the linear algebra is unpriced, and the rho reference is dominated by fixed set-up at these tiny `r` (11–67).
4. **The gap widens with p.** The least-squares log-log slope in p is 0.85 for IC on genus 2 and 1.06 for IC on genus 3, against 0.44 for rho on E (derived; 8 points, no interval). This is the `Ω(p)` relation count against rho's `√r`.
5. **Rho is unchanged by the transfer.** On the order-`r` subgroup of either Jacobian, rho matches rho on E: the subgroup is the same size.

Limits

- `p < 256` (`HCurve`) and `r ≤ 67`. These are toy sizes: rho is dominated by set-up and the exponents are fitted over a short range. Genus-2 IC at larger p is in `../RESULTS.md` (p up to 1619, 9.8–42.9× rho).
- One instance per prime. Spread is over seeds only.
- The cross map to the genus-3 curve's complementary factor (a genus-2 Jacobian, `w² = u·F(u)`) is zero by the same deck-involution argument as in `../ATLAS.md`. That argument is not measured here.

- No independent validation. The result is not admitted to the benchmarks table (validation gate).

Reproduce

```
cargo test  --release --lib chained_transfer::tower
cargo run   --release --bin tower_bench -- run --json tower.json --svg
tower.svg
```